Do not start attempting the test paper until you are asked to do so.

SILVER ZONE FOUNDATION
NEW DELHI, INDIA

ENROLLMENT NUMBER:

STUDENT NAME:

Class 6

Maximum Marks : 100
Duration : 40 Minutes
Total Questions : 50

Test Paper Type
AD

Level - 2

iOS 11th International Olympiad of Science 2019-2020

Instructions for the Candidates

1. Write your 12 digit iOS Enrollment Number and your name on top of the Question Paper in the given space.
2. Filling up of improper Enrollment Number may lead to unavailability of Result.
3. This Question Paper cum Answer Sheet must be submitted to your teacher after the completion of the test.
4. Tick ☑ your answer (A, B, C or D) on the given box ☐ in this Question Paper cum Answer Sheet with HB Pencil or Black/Blue Ball point Pen.
5. The Question Paper cum Answer Sheet Booklet consists of 50 questions, divided into three sections.
6. Section-A: Science (30 questions); Section-B: Reasoning and Aptitude (10 questions); Section –C: Scholar's Zone (10 questions).
7. Each question of Section-A carries 1.5 marks; Section-B carries 2.5 marks and that of Scholar's Zone carries 3 marks.
8. Use of calculator or any other devices in the examination is strictly prohibited.
9. Rough work should be done on the blank sheet/space provided in the question paper booklet.
10. All the questions are compulsory and there is no negative marking.

SIGN./NAME OF THE STUDENT

SIGN. OF THE TEACHER

1. The characteristics given below are of which one of the following diseases?

   I. This disease is caused by the deficiency of iodine.

   II. In children, this disease is known as the cretinism.

   III. The main symptom of this disease is that the neck of the person appears to be swollen.

   IV. It is common in hilly areas because the 'drinking water' and 'food' grown in such areas is poor in iodine.

   A. ☐ Rickets
   B. ☐ Scurvy
   C. ☐ Beriberi
   D. ☐ Goitre

2. In a test for the presence of protein in milk when we add 2 to 3 drops of copper sulphate solution on a milk mixed with dilute sodium hydroxide solution, which colour is produced?

   A. ☐ Yellow
   B. ☐ Red
   C. ☐ Violet
   D. ☐ Pink

3. In the picture given below, the child is suffering from _____, which is a _____ deficiency disease.



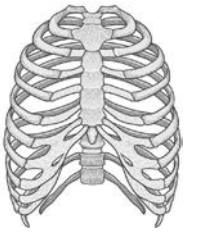
   A. ☐ marasmus, iron
   B. ☐ kwashiorkor, protein
   C. ☐ anaemia, carbohydrate
   D. ☐ None of these

4. The process shown by the picture given below is:



   A. ☐ Weaving
   B. ☐ Spinning
   C. ☐ Ginning
   D. ☐ Knitting

5. The structure shown below is found in the human body. Choose the incorrect statement about the structure.



   A. ☐ It protects the internal organs such as heart and lungs.
   B. ☐ It protects a part of the stomach and the kidneys.
   C. ☐ It provides a surface for the attachment of muscles concerned with the respiratory mechanism.
   D. ☐ None of these

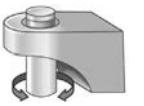
6. The fibres which obtained from plants and animals are called _____ fibres, e.g, _____ and _____.

   A. ☐ synthetic, nylon, acrylic
   B. ☐ synthetic, cotton, flax
   C. ☐ natural, rayon, nylon
   D. ☐ natural, cotton, silk

7. Which part of the cotton plant is used for getting cotton fibre?

   A. ☐ Roots
   B. ☐ Stem
   C. ☐ Leaves
   D. ☐ Fruit

8. Which of the following represents a ball and socket joint?

   A. ☐ 
   B. ☐ 
   C. ☐ 
   D. ☐ 

9. The blue garbage bin is used to throw:

   A. ☐ Banana peels
   B. ☐ Rotten eggs
   C. ☐ Leftover cooked food
   D. ☐ Plastic wares, Glass jars and bottles

10. The acronym of BOD is:

    A. ☐ biological oxygen demand
    B. ☐ bacterial oxygen demand
    C. ☐ biological oxide demand
    D. ☐ biological ozone demand

11. Which one of the following is correct about the magnetic field?

A. ☐ A magnetic field is the region around a magnet in which only force of attraction can be experienced.

B. ☐ A magnetic field is the region around a magnet in which only force of repulsion can be experienced.

C. ☐ A magnetic field is the region around a magnet in which magnetic force is experienced by a magnetic material.

D. ☐ All of these

12. Devices that are used to convert electrical energy into heat energy are:

A. ☐ Television and Radio

B. ☐ Iron and Bulb

C. ☐ Geyser and Heater

D. ☐ Mixer and Toaster

13. Which one of the following parts of our body does not contain cartilage?

A. ☐ Pinnae of ear

B. ☐ End of nose

C. ☐ Ear lobe

D. ☐ Between the vertebrae of the backbone

14. The height of a boy is 5 metres and 2 centimetres. Convert his height into decimetre.

A. ☐ 1.55 dm

B. ☐ 50.2 dm

C. ☐ 152 dm

D. ☐ 15.9 dm

15. Which one of the following is a gastropod?

A. ☐ 

B. ☐ 

C. ☐ 

D. ☐ Both A and C

16. A train travels from Delhi with a velocity of 70 km/h northwards. What is its displacement after 5 hours?

A. ☐ 150 km north

B. ☐ 250 km north

C. ☐ 350 km north

D. ☐ 450 km north

17. _____ is not always necessary to observe a shadow.

A. ☐ Sun

B. ☐ Screen

C. ☐ Source of light

D. ☐ Opaque object

18. Our Sun is a part of which of the following galaxies?

A. ☐ Andromeda

B. ☐ Orion

C. ☐ Milky way

D. ☐ Comet

19. Which one of the following flowers is not eaten as food?

A. ☐ Sunflower

B. ☐ Dandelion

C. ☐ Gladiolus

D. ☐ None of these

20. The figure given below is showing which of the following kinds of separation?



A. ☐ Threshing

B. ☐ Sieving

C. ☐ Filtration

D. ☐ Winnowing

21. Liquid waste collected from houses is known as:

A. ☐ Effluent

B. ☐ Smoke

C. ☐ Smog

D. ☐ Sewage

22. Rain water harvesting means:

A. ☐ harvesting of crops during rains.

B. ☐ using rain water to add to underground water.

C. ☐ storing rain water in tanks.

D. ☐ Both B and C

23. In which of the following layers of the atmosphere most of the ozone is present?

A. ☐ Troposphere

B. ☐ Exosphere

C. ☐ Mesosphere

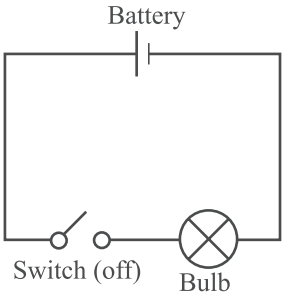
D. ☐ Stratosphere

24. An object, 5 m high, is placed at a distance of 8 m in front of a plane mirror. Find the distance between the object and its image.

A. ☐ 5 m B. ☐ 8 m

C. ☐ 13 m D. ☐ 16 m

25. In an electric circuit, the conventional direction of flow of electric current is taken to be from:



A. ☐ positive terminal to negative terminal.
B. ☐ negative terminal to positive terminal.
C. ☐ negative terminal to neutral terminal.
D. ☐ positive terminal to neutral terminal.

26. Which one of the following mirrors is used by a dentist to examine teeth?

A. ☐ Simple mirror
B. ☐ Concave mirror
C. ☐ Convex mirror
D. ☐ All of these

27. Which one of the following statements is incorrect?

A. ☐ Statement 1: Sea water is a mixture of water, common salt and various other salts.
B. ☐ Statement 2: Crude oil is a mixture of various carbon compounds.
C. ☐ Statement 3: Air is a mixture of many gases.
D. ☐ None of these

28. Which vitamin is easily destroyed by heat during the cooking of food?

A. ☐ Vitamin A B. ☐ Vitamin C
C. ☐ Vitamin D D. ☐ Vitamin B

29. Which one of the following is a parasitic plants?

A. ☐ Rose plant B. ☐ Jasmine plant
C. ☐ Mango plant D. ☐ Cuscuta

30. Eating too much of fat rich food can lead to a disease called _____.

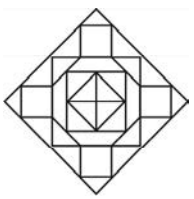
A. ☐ marasmus B. ☐ obesity
C. ☐ cretinism D. ☐ kwashiorkor

Section–B (Reasoning and Aptitude)

31. Mr. Das left for his office in his car. He drove 12 km towards North and then 10 km towards West. He then turned to the South and covered 4 km. Further, he turned to the East and moved 8 km. Finally, he turned right and drove 8 km. How far and in which direction is he from his starting point?

A. ☐ 2 km, West B. ☐ 2 km, East
C. ☐ 4 km, North D. ☐ 2 km, South

32. How many squares are there in the given figure?

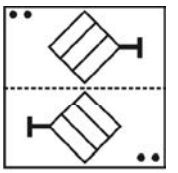


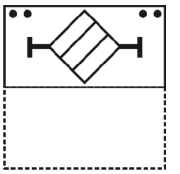
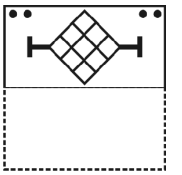
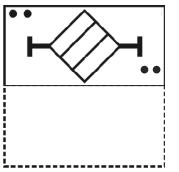
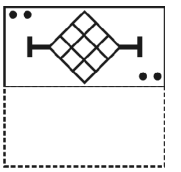
A. ☐ 9 B ☐ 10
C. ☐ 11 D. ☐ 12

33. If 'Book' is called 'Pen', 'Pen' is called 'Cell' and 'Cell' is called 'Copy', then we write with a _____.

A. ☐ Book B. ☐ Pen
C. ☐ Copy D. ☐ Cell

34. A transparent sheet (X) with a design and a dotted line on it is given below.
Choose the correct figure from the four options that is the folded form of the Sheet (X).


Sheet (X)

A. ☐ 
B. ☐ 
C. ☐ 
D. ☐ 

35. In a certain code RESPIRATION is as TGUQKTCVKQP, how will MILITARY be written in that code?

A. ☐ LNKHSZQX B. ☐ NJMJSBSZ
C. ☐ OKNKVCTA D. ☐ OKNKTCTP

36. Pointing to a lady in a photograph Rinku said, "She is the mother of my mother's husband". How is the lady related to Rinku?

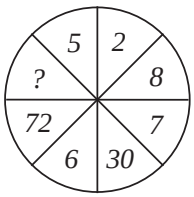
A. ☐ Mother
B. ☐ Aunt
C. ☐ Sister
D. ☐ Grandmother (paternal)

37. Which of the following word can be formed by using the letters of the word given below.

RATIONALISATION

A. ☐ NATIONALIST
B. ☐ NATIONALISTIC
C. ☐ REALISATION
D. ☐ SITUATION

38. Find the missing number:



A. ☐ 54 B. ☐ 49
C. ☐ 63 D. ☐ 56

39. If + stand for ×, × stands for ÷, ÷ stands for – and – stands for +, then 2 – 8 × 2 + 6 ÷ 7 = ?

A. ☐ 32 B. ☐ 19
C. ☐ 23 D. ☐ 12

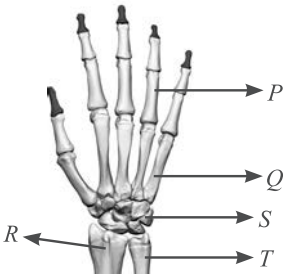
40. Identify the relationship between the pair of two words given on either side of :: and then find the missing word.

Haemoglobin : Iron :: Chlorophyll : ?

A. ☐ Copper B. ☐ Magnesium
C. ☐ Cobalt D. ☐ Calcium

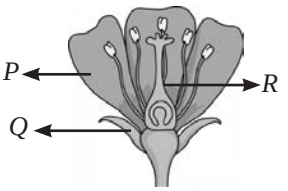
Space for Rough Work

41. Choose the correct option to label the diagram given below.



A. ☐ P → Radius, Q → Metacarpals, R → Carpals, S → Phalanges, T → Ulna
B. ☐ P → Phalanges, Q → Metacarpals, R → Ulna, S → Carpals, T → Radius
C. ☐ P → Phalanges, Q → Metacarpals, R → Radius, S → Carpals, T → Ulna
D. ☐ P → Metacarpals, Q → Metacarpals, R → Radius, S → Carpals, T → Ulna

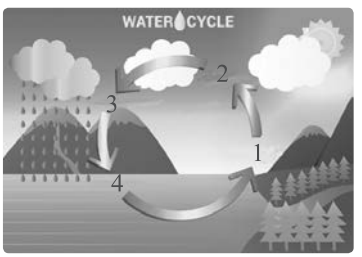
42. Which combination of the statements is correct for the picture given below?



i. *The flower is that part of a plant which contains the reproductive organs and produces fruits and seeds.*
ii. *P is known as petal.*
iii. *P is the most attractive part of a flower. It can be of different colours, shapes and sizes.*
iv. *R is the female part of a flower and is made up of three parts: stigma, style and ovary and is also known as the 'carpel'.*
v. *Q is the male part of the flower and contains many little stalks with swollen tops.*

A. ☐ i, iii and v
B. ☐ i, ii, iii and iv
C. ☐ ii, iii and iv
D. ☐ i, ii and iii

43. Arrange the processes involved in the water cycle in correct sequence on the basis of the figure given below.



A. ☐ Evaporation, Condensation, Precipitation, Accumulation
B. ☐ Condensation, Evaporation, Precipitation, Accumulation
C. ☐ Evaporation, Condensation, Accumulation, Precipitation
D. ☐ Precipitation, Evaporation, Condensation, Accumulation

44. Match the following:

	Column A		Column B
i.	*Collar bone*	*a.*	*Ulna*
ii.	*Forearm bone*	*b.*	*Femur*
iii.	*Thigh bone*	*c.*	*Clavicle*
iv.	*Lower leg bone*	*d.*	*Tibia*

A. ☐ i → d, ii → c, iii → a, iv → b
B. ☐ i → c, ii → a, iii → b, iv → d
C. ☐ i → b, ii → c, iii → d, iv → a
D. ☐ i → c, ii → d, iii → b, iv → a

45. Raman gave the following ill effects of the practice of burning dried leaves and other plant products.

i. *Burning of dried leaves produces precious manures.*
ii. *Burning produces harmful gases/fumes.*
iii. *Precious raw materials to obtain manure at low cost is lost.*
iv. *Unnecessarily lot of heat is generated.*

The correct reasons for why we should not burn leaves are:

A. ☐ i, ii and iii only
B. ☐ ii, iii and iv only
C. ☐ i and iv only
D. ☐ i, ii, iii and iv

46. Gaurav took an empty plastic bottle, turned it upside down and dipped its open mouth into a bucket filled with water. He then tilted the bottle slightly and made the following observations.

i. *Bubble of air came out from the bottle.*
ii. *Some water entered the bottle.*
iii. *Only nitrogen gas came out in the form of bubbles and oxygen got dissolved in water.*
iv. *No bubbles formed, only water entered the bottle.*

Which observations is/are correct?

A. ☐ i and ii
B. ☐ iv only
C. ☐ iii and iv
D. ☐ i only

47. Fill in the blanks with the correct combination of words.

___L___ is the chemical process in which ___M___ taken by an organism combines with ___N___ to release ___O___ .

A. ☐ L → Nutrition, M → food, N → water, O → energy

B. ☐ L → Respiration, M → food, N → oxygen, O → energy

C. ☐ L → Respiration, M → food, N → nitrogen, O → glucose

D. ☐ L → Photosynthesis, M → carbon dioxide, N → oxygen, O → energy

48. Match the following:

	Column A		Column B
i.	*The movement of a car on circular path*	*a.*	*Rectilinear motion*
ii.	*The motion of a man on straight path*	*b.*	*Rotational motion*
iii.	*The motion of stretched strings of guitar*	*c.*	*Circular motion*
iv.	*The motion of a spinning top*	*d.*	*Vibratory motion*

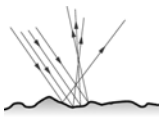
A. ☐ i → c, ii → d, iii → a, iv → b

B. ☐ i → c, ii → a, iii → b, iv → d

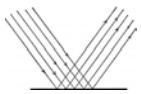
C. ☐ i → a, ii → b, iii → c, iv → d

D. ☐ i → c, ii → a, iii → d, iv → b

49. Look at the following figures and choose the correct option.



X



Y

i. *X shows reflection by a rough surface.*

ii. *X shows the phenomenon of regular reflection.*

iii. *Y shows reflection by a smooth surface.*

iv. *Y shows the phenomenon of an irregular reflection.*

A. ☐ i and ii B. ☐ ii and iii

C. ☐ iii and iv D. ☐ i and iii

50. Which combination of the statements is correct?

Statement 1: The arrangement of veins in the lamina is known as the venation of the leaf.

Statement 2: The veins in a leaf occur in an irregular way forming a net like design is called the parallel venation.

Statement 3: The veins in a leaf run parallel to one another on both the sides of the mid rib is called the reticulate venation.

Statement 4: The plants having leaves with parallel venation have fibrous roots.

A. ☐ 2, 3 and 4 only

B. ☐ 1 and 4 only

C. ☐ 2 and 3 only

D. ☐ All of these

Space for Rough Work

ANSWERS

INTERNATIONAL OLYMPIAD OF SCIENCE 2019 (LEVEL 2) - CLASS 6

1.	D	2.	C	3.	B	4.	C	5.	D	6.	D	7.	D	8.	A	9.	D	10.	A
11.	C	12.	C	13.	C	14.	B	15.	A	16.	C	17.	A	18.	C	19.	D	20.	B
21.	D	22.	D	23.	D	24.	D	25.	A	26.	B	27.	D	28.	B	29.	D	30.	B
31.	A	32.	D	33.	D	34.	B	35.	C	36.	D	37.	A	38.	D	39.	B	40.	B
41.	C	42.	B	43.	A	44.	B	45.	B	46.	A	47.	B	48.	D	49.	D	50.	B